

Discrete

A Post-Quantum Proof-of-Work Cryptocurrency

The Discrete developers

discrete.cash

Pre-launch protocol paper, revision 1.0 — July 2026

Candidate consensus constants; independent review pending

Abstract

Discrete is a pre-launch proof-of-work cryptocurrency whose transaction validation is post-quantum from genesis. Ownership uses ML-KEM-768 for private output delivery and ML-DSA-65 for spend authorization, with deterministic seed recovery, one-time output credentials, and nullifier-based double-spend prevention. Amounts and the spent-output graph are transparent in Phase 1. Mining uses DiscretePower-1: a SHAKE-256 transcript in which every nonce is signed by the coinbase identity's ML-DSA key before the yespower 1.0 memory-hard core is evaluated. This binds work and reward identity and complicates conventional delegation, but it is not claimed to be a formally strong non-outsourceable puzzle. Each node also rejects reorganizations deeper than ten blocks relative to its local accepted history; this weak-subjective rule protects already-connected nodes from late private rewrites at the cost of partition wedges and operator-guided recovery. Issuance follows a decaying exponential curve into a perpetual approximately 2%-per-year tail. This paper specifies the candidate protocol, wire formats, monetary policy, recovery procedure, residual risks, and the limits that remain before public launch and independent review.

Genesis anchor: "Reuters 08/Jul/2026 — Crypto firms prepare defenses as quantum threat to encryption draws nearer"

Contents

1	Introduction	3
1.1	The quantum threat to existing cryptocurrencies	3
1.2	Post-quantum from genesis	3
1.3	Beyond cryptography: the consensus layer	3
1.4	Organization and status	4
2	Design principles	4
3	Cryptographic foundations	5
3.1	Primitive suite	5
3.2	Notation	5
3.3	Domain separation	6
3.4	The cost of post-quantum sizes	6
4	Keys, accounts, and addresses	6
4.1	Deterministic key derivation from one seed	6
4.2	Address format	7
4.3	On-chain account numbers	7
4.4	Unlimited deposit addresses from one key	8
5	Transactions	8
5.1	Output construction	8
5.2	The ownership model, and why it has a long-term spend key	10
5.3	Scanning and wallet recovery	10
5.4	Spending, the signing digest, and nullifiers	11
5.5	Consensus validation rules	12
5.6	Canonical denominations	12
5.7	Per-output time-locks	12
5.8	Fees	13
5.9	Payment proofs	13
5.10	Transaction sizes	13
6	Consensus	14
6.1	Identity-bound signed proof-of-work	14
6.2	Coinbase bound to the block signer	15
6.3	Difficulty adjustment	16
6.4	Local first-seen depth limit	16
6.5	Why the reference implementation includes no decentralized pool	17
6.6	Network layer	18
7	Monetary policy	18
7.1	The emission function	18
7.2	Two phases	18
7.3	Why a perpetual tail	18
7.4	Units and parameters	19
7.5	The Treasury Reserve: transparent and vested	19

8	Security analysis	20
8.1	Threat model	20
8.2	Ownership and supply integrity	20
8.3	Attack catalogue	21
8.4	Consensus attacks	21
8.5	Costs stated plainly	22
9	Roadmap: staged discretion	22
9.1	Confidential amounts	22
9.2	On untraceability	23
10	Related work	23
11	Conclusion	23
A	Candidate network parameters	23
B	Domain-separation tags	24
C	Wire formats	25

1 Introduction

1.1 The quantum threat to existing cryptocurrencies

Bitcoin authorizes spending with ECDSA over secp256k1; most chains built since use Ed25519 or close relatives. The security of every one of these schemes reduces to the hardness of the elliptic-curve discrete-logarithm problem, which is secure against all known classical algorithms and *completely broken* by a sufficiently large quantum computer running Shor's algorithm [1]: given a public key, the algorithm recovers the private key in polynomial time. On a cryptocurrency this directly threatens spend authorization and any classically signed control plane:

- **Ownership is forgeable.** Any address whose public key has been exposed — every address that has ever spent, and on many chains every address outright — can be drained by an adversary who forges its signatures.
- **Classically signed control planes are forgeable.** A chain that gives checkpoint, bridge, treasury, or governance authority to an elliptic-curve key exposes that authority as well. This is distinct from proof-of-work subsidy validation, which does not ordinarily depend on an ownership signature.
- **The exposure persists.** An adversary can archive exposed public keys today and forge ownership signatures if suitable hardware later matures. This is a future-forgery risk, not decryption of encrypted chain data.

The industry's dominant response is to plan *migrations*: bolting post-quantum signature schemes onto chains that were born classical. Migrations of live monetary systems are slow and contentious, require flag days, and inevitably leave a long tail of un-migrated outputs — coins whose owners are inactive, deceased, or unaware — that remain permanently vulnerable and whose eventual theft or freezing is a governance crisis waiting to happen.

1.2 Post-quantum from genesis

Discrete takes the other path. It is post-quantum **from genesis**: there is no elliptic-curve operation anywhere in its transaction validation path, no vulnerable legacy state, no migration plan because there is nothing to migrate. Every key, signature, and output from block 0 onward is built on the algorithms the U.S. National Institute of Standards and Technology standardized for exactly this threat: ML-KEM-768 (FIPS 203 [2]) and ML-DSA-65 (FIPS 204 [3]), both at NIST Security Category 3.

The substantive technical achievement is not merely a post-quantum transparent transaction format. It is the reconstruction of CryptoNote-style stealth ownership [8] — static reusable addresses, per-output one-time credentials, deterministic recovery of an entire wallet from one seed, and nullifier-based double-spend prevention — using *only* standardized NIST post-quantum primitives and symmetric cryptography (Section 5).

1.3 Beyond cryptography: the consensus layer

Post-quantum keys defend against tomorrow's adversary; a young proof-of-work chain must first survive today's. The attacks that have actually defeated small proof-of-work coins are not cryptanalytic — they are *hashrate-assembly* attacks: renting hashrate, incentivizing miners to point at a hostile pool, and using the assembled majority to rewrite confirmed history. In 2025 this reached even the largest ASIC-resistant chain in existence, whose

history was deeply reorganized by an external pool that simply out-bid the honest network for rented and incentivized hashrate [17].

Discrete’s consensus design (Section 6) starts from the principle that *51% resistance cannot come from inside proof-of-work*: any rule verifiable purely from hashrate is satisfied by whoever holds the most of it. Discrete therefore deploys two mechanisms:

1. **Identity-bound signed proof-of-work** (SPoW, Section 6.1): every unit of valid work is an ML-DSA signature-then-hash bound to the key that controls the block reward. Every nonce needs the reward identity’s signature. This blocks unsigned reward redirection and adds a signing bottleneck to delegation, but a custodial operator can still sign and distribute candidate jobs. The mechanism descends from Karbo; operational history is not a formal non-outsourcability proof.
2. **Node-local first-seen depth limit** (Section 6.4): each node records which chain it witnessed growing and refuses any reorganization deeper than a fixed finality depth, however much work the competing chain carries. Observation order is a resource an attacker cannot obtain by having more hashrate.

1.4 Organization and status

Section 2 states the design principles. Section 3 fixes the cryptographic suite and notation. Sections 4 and 5 specify accounts, addresses, and the transaction family with pseudocode matching the reference implementation. Section 6 specifies consensus: the signed proof-of-work, difficulty adjustment, and the finality rule with its recovery procedure. Section 7 specifies monetary policy and the genesis Treasury Reserve. Section 8 analyzes security and states the design’s costs. Section 9 describes the staged path to confidential amounts. Appendices A–C collect the candidate network parameters, domain-separation tags, and wire formats.

This document is a pre-launch technical overview drafted against candidate consensus constants. The implementation, specification files, and pinned known-answer vectors must be reconciled byte-for-byte before a launch tag; the public release must identify the exact repository commit reviewed.

2 Design principles

Discrete is deliberately *compact and auditable*. The reference implementation is a pre-launch, post-quantum, transparent-amount chain, with confidential-amount discretion staged as a separately reviewed and independently audited fork (Section 9). Four principles govern every choice in this paper:

Standardized primitives only in consensus. Discrete uses NIST-standardized post-quantum algorithms (FIPS 203, FIPS 204) and well-established symmetric primitives (SHA-3 [4], HKDF [5], ChaCha20-Poly1305 [6]). No novel, un-cryptanalyzed construction sits on the critical path.

Inherited mechanisms, re-evaluated. Parts of the mining design descend from Karbo, but operational history is evidence rather than a proof. Discrete states separately what identity-bound work, local finality, and recovery do and do not guarantee.

Objective rules separated from local state. Emission and ownership are functions of public consensus data. The depth-10 reorganization limit is relative to a node’s locally

accepted history, and exceptional recovery uses an operator-confirmed out-of-band majority signal.

Honest about what is and is not protected. The monetary policy has no hard cap (by design, Section 7); mining is CPU-egalitarian and therefore not botnet-proof (Section 6.1); finality trades a recoverable liveness events for protection of already-observed deep history (Section 6.4); and Phase-1 spends are publicly linkable (Section 5.2). Each cost is deliberate and documented.

3 Cryptographic foundations

3.1 Primitive suite

Discrete’s consensus uses exactly the primitives in Table 1. Both lattice schemes are NIST Security Category 3, a strength target comparable to AES-192, against both classical and quantum adversaries.

Table 1: Normative cryptographic suite.

Role	Primitive	Standard	Sizes
View / scan key	ML-KEM-768	FIPS 203	pk 1,184 B; sk 2,400 B; ct 1,088 B; ss 32 B
Spend authority	ML-DSA-65	FIPS 204	pk 1,952 B; sk 4,032 B; sig 3,309 B; seed 32 B
Hash	SHA3-256	FIPS 202	output 32 B
KDF	HKDF-SHA3-256	RFC 5869 + FIPS 202	salt 0^{32} ; L as specified
AEAD	ChaCha20-Poly1305	RFC 8439 (IETF)	key 32 B; nonce 12 B; tag 16 B
Proof-of-work	DiscretePower-1	FIPS 202/204 + yespower 1.0	8 MiB scratchpad
Address encoding	extended Bech32m-derived	custom length profile	HRP disc / tdisc

Two conventions apply throughout:

- **HKDF instantiation.** Every HKDF call is HKDF-SHA3-256 with a fixed salt of 32 zero bytes; the *info* string is the domain tag shown in each derivation, and the output length is $L=32$ unless stated otherwise. Implementations must not vary salt or length.
- **AEAD nonce.** The ChaCha20-Poly1305 nonce is 12 zero bytes. This is safe because every AEAD key is unique per output — it is derived from a per-output KEM shared secret — and key uniqueness makes nonce reuse structurally impossible rather than merely improbable.

3.2 Notation

$H(\cdot)$ denotes SHA3-256. $a \parallel b$ denotes byte-string concatenation. $\text{LE}_{32}(x)$ and $\text{LE}_{64}(x)$ denote 32- and 64-bit little-endian encodings. Domain tags are ASCII strings hashed *without* a trailing NUL. $\text{HKDF}(\text{ikm}, \text{info}, L)$ abbreviates HKDF-SHA3-256($\text{IKM}=\text{ikm}$, $\text{salt}=0^{32}$, info , L). For the KEM, $(ct, ss) \leftarrow \text{ML-KEM-768.Encaps}(pk)$ and $ss \leftarrow \text{ML-KEM-768.Decaps}(sk, ct)$; for the signature scheme, $(pk, sk) \leftarrow \text{ML-DSA-65.KeyGen}(\xi)$ with 32-byte seed ξ , $\sigma \leftarrow \text{ML-DSA-65.Sign}(sk, m)$, and $\text{ML-DSA-65.Verify}(pk, m, \sigma) \in \{0, 1\}$.

3.3 Domain separation

All key material and every consensus-critical hash in Discrete is derived under a distinct, versioned ASCII domain tag with the prefix `discrete-`, so that (i) material derived for one purpose can never be reused for another, and (ii) material derived for Discrete can never collide with another system — including its own design ancestor, whose tags used the prefix `karbo-pq-`. The full normative table appears in Appendix B; tags reserved for future phases (`discrete-pq-ct-mask-v1`, `discrete-pq-serial-v1`) must not be used by Phase-1 code, and unit tests pin every tag byte-for-byte. A mismatch in any derivation between independent implementations is a chain split; known-answer test vectors for every derivation are published with the source.

3.4 The cost of post-quantum sizes

Lattice keys and signatures are one to two orders of magnitude larger than the elliptic-curve material they replace, and Discrete accepts this cost explicitly rather than compressing it away with non-standard tricks. The candidate consensus sizes are pinned in code and tests: a spend reveals a 1,952-byte ML-DSA-65 public key and a 3,309-byte signature; an output carries a 1,088-byte ML-KEM-768 ciphertext. These sizes drive several engineering choices documented later: coinbase outputs omit the KEM ciphertext entirely and are recovered deterministically (Section 6.2), saving 1,144 bytes per block; fees are flat rather than size-proportional (Section 5.8); and human-facing identity uses short on-chain account numbers rather than raw addresses (Section 4.3).

4 Keys, accounts, and addresses

4.1 Deterministic key derivation from one seed

A Discrete account is a pair of independent post-quantum keys derived from a single 32-byte master seed (BIP-39 mnemonic or CSPRNG output). The two branches are domain-separated so that neither key can be inferred from the other or from the seed of the other branch:

$$view_seed = \text{HKDF}(seed_{\text{master}}, \text{discrete-pq-view-root-v1}, L=64) \quad (1)$$

$$(vpk, vsk) = \text{ML-KEM-768.KeyGen}(view_seed) \quad (2)$$

$$spend_seed = \text{HKDF}(seed_{\text{master}}, \text{discrete-pq-spend-root-v1}, L=32) \quad (3)$$

$$(spk, ssk) = \text{ML-DSA-65.KeyGen}(\xi = spend_seed) \quad (4)$$

The view branch uses $L=64$ because FIPS 203 key generation consumes 64 bytes ($d \parallel z$); the spend branch uses $L=32$ because FIPS 204 takes a 32-byte seed ξ . The **view key** (vpk, vsk) receives and audits: it decapsulates the per-output delivery ciphertexts and can therefore see everything the account receives, but cannot spend. The **spend key** (spk, ssk) is the sole spending authority. Backing up the one master seed reconstructs the full post-quantum identity; recovery is a deterministic chain rescan (Section 5.3) with no per-output key material ever stored.

4.2 Address format

A Discrete address carries both public keys, so that any sender can deliver an output (via *vpk*) and bind it to the recipient's spending authority (via *spk*):

Field	Content	Size
version	= 1	1 B
networkPrefix	varint network discriminator	1–2 B
viewPub	ML-KEM-768 public key	1,184 B
spendPub	ML-DSA-65 public key	1,952 B
checksum	$H(\text{version} \parallel \text{prefix} \parallel \text{vpk} \parallel \text{spk})[0:4]$	4 B

The serialized address uses the **Bech32m alphabet and polymod constant** [14], plus an inner four-byte SHA3 checksum. It is not a standard BIP-350 Bitcoin address: BIP-173's generic Bech32 profile caps strings at 90 characters [15], while a Discrete base address is approximately 5,043 characters. Addresses are self-identifying by network: the main-net human-readable prefix is `disc` (addresses render as `disc1...`) and testnet is `tdisc` (`tdisc1...`), so the two can never be confused; a wallet rejects an address from the other network even though the underlying keys would parse. Because the payload is over 3 KB, a rendered address runs to several thousand characters — intended for files, payment URIs, or copy–paste, not transcription. It exceeds the capacity of a single standard QR symbol; multi-frame transport would require a separate framing protocol. Human-facing identity is handled by the account-number layer below.

4.3 On-chain account numbers

Discrete adds an optional on-chain **account-number registry** that maps short, human-checkable identifiers to full addresses. A user registers once by publishing a transaction whose `tx_extra` carries a registration tag binding the identity's two public keys:

$$\text{registration tag} = \text{tag} \parallel \text{vpk} (1,184 \text{ B}) \parallel \text{spk} (1,952 \text{ B}).$$

Consensus enforces *first-registration-wins*: the first confirmed transaction registering a given key pair is canonical, and any later attempt is invalid. The account number is not chosen by the user; it is derived from the registration's position in the chain — the pair (*blockHeight*, *txIndex*) — making it immutable, globally unique, and resolvable by any node with no extra trust. It renders in **H-I-C** form,

$$\underbrace{\text{height}}_H - \underbrace{\text{txIndex}}_I - \underbrace{c}_{\text{check}}$$

where *c* is a Luhn mod-36 check character over the concatenated decimal digits of *H* and *I*, so any single transcription typo is caught locally before a lookup. Registration persists in the node database under two indices (key → position for duplicate rejection; position → key for resolution) and rolls back across reorgs exactly like the nullifier set (Section 5.4).

Free registration for new users. A new user holds no coins and cannot pay a fee, but needs an on-chain registration to be practically payable. Discrete resolves this bootstrap problem with a dedicated zero-fee transaction type, `TX_FREE_REG`, whose anti-spam cost is a memory-hard proof-of-work instead of a fee. A free-registration transaction is valid iff it

has no inputs and no outputs, carries exactly one registration tag and one PoW tag, and satisfies

$$\text{yespower}(\text{discrete-pq-free-reg-pow-v1} \parallel \text{vpk} \parallel \text{spk} \parallel \text{refBlockHash} \parallel \text{LE64}(\text{nonce}))_{[0:8]} \leq \text{FREE_REG_POW_TARGET}. \quad (5)$$

where both public keys are bound so one proof cannot be replayed with different spend identities, and *refBlockHash* must be a main-chain block within the last `FREE_REG_REF_WINDOW` = 60 blocks (preventing precomputation hoarding) and the target `0x00007FFFFFFFFFFFFF` implies an expected $2^{17} = 131,072$ yespower evaluations — with the 8-MiB block-PoW parameters, minutes of one-time CPU work for a legitimate user, but real CPU-hours for bulk squatting. Blocks admit at most `FREE_REG_PER_BLOCK` = 100 such transactions, and the mempool caps pending registrations at 2,000 so zero-fee traffic can never crowd out paid transactions.

4.4 Unlimited deposit addresses from one key

Exchanges and merchants need a distinct address per customer or invoice. Classical chains answer with subaddress schemes that multiply scan work or key state; Discrete instead binds a 64-bit **routing index** T into each output’s delivery. T is not a key: spend authority is unchanged, and the index is recovered from the output itself during normal decapsulation — no lookup table, no extra scan pass. A deposit identifier renders as **H-I-T-C** (the base account plus index plus a Luhn check character over all three fields), which can never alias the two-dash base form. One account can therefore hand every counterparty its own deposit address and recover all of them from the single master seed.

For deployments that prefer distinct spend keys per deposit, the wallet layer additionally supports an *aggregated-multikey* family: per-deposit ML-DSA-65 spend keys

$$\begin{aligned} \xi_i &= \text{HKDF}(\text{seed}_{\text{master}}, \text{discrete-pq-deposit-spend-v1} \parallel \text{LE32}(i), 32), \\ (\text{spk}_i, \text{ssk}_i) &= \text{ML-DSA-65.KeyGen}(\xi_i), \end{aligned} \quad (6)$$

all sharing the single view key, so scanning still costs one decapsulation per output. Both derivations are cemented: changing the domain, the index encoding, or the HKDF parameters would strand existing deposits on recovery.

5 Transactions

Discrete has one transaction version and four transaction types: `TX_COINBASE` (0x00, block reward), `TX_PQ` (0x01, ordinary transfer), a permanently reserved type 0x02, and `TX_FREE_REG` (0x03, Section 4.3). This section specifies the life of a `TX_PQ` output: construction, delivery, scanning, spending, and validation.

5.1 Output construction

Per-output randomness. Every output carries a secret 32-byte blinding value ρ , drawn fresh from a CSPRNG by the sender. ρ is delivered encrypted to the recipient and revealed only when the output is spent; it binds the output’s credentials to that specific output so no other output’s material can be substituted.

Consensus-enforced output uniqueness. A stable per-output identifier, *outContext*, is computed before any key derivation. Uniqueness must be *enforced*, not assumed: a malicious sender who could replay the same KEM ciphertext and output index in two transactions would produce two outputs with identical credentials, of which only one could ever be spent. Discrete folds the transaction’s *inputs hash* into the context; since consensus guarantees a given set of UTXOs is spent at most once, the inputs hash is a uniqueness source that costs no new fields:

$$inputsHash = H\left(\text{discrete-pq-inputs-hash-v1} \parallel (prevTxid_j \parallel \text{LE32}(prevOutIndex_j))_{j=1..n}\right) \quad (7)$$

$$outContext = H\left(\text{discrete-pq-out-context-v1} \parallel inputsHash \parallel kemCt \parallel \text{LE32}(i) \parallel \text{LE64}(T)\right) \quad (8)$$

where *i* is the output’s index in the transaction and *T* is the routing index of Section 4.4 (*T*=0 for ordinary payments).

Delivery, binding, and authenticated encryption. The sender encapsulates to the recipient’s view key, derives a one-time AEAD key, and seals ρ and *T*; the public amount is bound into the authentication tag as associated data, so any in-flight tampering with the amount, the context, or the routing hint makes the output invisible to the recipient rather than wrongly credited:

$$(kemCt, ss) \leftarrow \text{ML-KEM-768.Encaps}(vpk) \quad (9)$$

$$k_{\text{aead}} = \text{HKDF}(ss, \text{discrete-pq-aead-key-v1} \parallel outContext, 32) \quad (10)$$

$$encPayload = \text{AEAD.Enc}(k_{\text{aead}}, \text{nonce}=0^{12}, \text{aad}=outContext \parallel \text{LE64}(a), \rho \parallel \text{LE64}(T)) \quad (11)$$

yielding 40 bytes of ciphertext plus a 16-byte tag (56 bytes on the wire, consensus-enforced). Finally the output commits to the recipient’s spending authority:

$$spendCommit = H(\text{discrete-pq-spend-commit-v1} \parallel spk \parallel \rho). \quad (12)$$

The on-chain output is ($\text{LE64}(a)$, $\text{LE64}(\text{unlockHeight})$, $0x10$, *kemCt*, *encPayload*, *spendCommit*) — 1,193 bytes. Note what is *absent*: neither public key appears. Until the output is spent, an observer sees only a hash commitment and an encapsulation that is indistinguishable from random without *vsk* — this is Discrete’s post-quantum stealth receiving.

Garbage outputs. A malicious sender can emit an output whose fields are well-formed but meaningless. This is inherent to every stealth-address system: the node cannot detect it without the recipient’s view secret. The recipient’s wallet handles it correctly and silently — decapsulation of a garbage ciphertext returns a pseudorandom secret, the derived AEAD key is wrong, tag authentication fails, and the output is discarded before it ever appears in a balance. The only party harmed is the sender, whose coins are burned; the fee requirement makes this an expensive way to spam.

Algorithm 1 CREATEOUTPUT — sender side (per output i of amount a to address (vpk, spk) , routing index T).

```

function CreateOutput(inputsHash, i, a, unlockHeight, vpk, spk, T):
    rho ← CSPRNG(32)
    (kemCt, ss) ← ML-KEM-768.Encaps(vpk)
    ctx ← SHA3-256("discrete-pq-out-context-v1" || inputsHash || kemCt
                  || LE32(i) || LE64(T))
    k ← HKDF(ss, "discrete-pq-aead-key-v1" || ctx, 32)
    enc ← AEAD.Enc(k, nonce=012, aad=ctx || LE64(a), msg=rho || LE64(T))
    sc ← SHA3-256("discrete-pq-spend-commit-v1" || spk || rho)
    // sender self-check: run the receiver predicate (Alg. 2) on (kemCt, enc, sc)
    // using ss directly; abort on any mismatch
    return Output{ a, unlockHeight, 0x10, kemCt, encPayload=enc, spendCommit=sc }

```

5.2 The ownership model, and why it has a long-term spend key

Discrete’s design ancestor (the Karbowanec PQ proposal, “KDSK”) derived a *per-output* spend key directly from the KEM shared secret:

$$\zeta_i = \text{HKDF}(ss, \cdot), \quad (pk_i, sk_i) = \text{ML-DSA-65.KeyGen}(\zeta_i).$$

This is elegant — no spend key in the address, per-output keys for free — and **unsound**: a KEM establishes the *same* shared secret for both parties, and the *sender* runs Encaps and therefore knows ss . Since sk_i is a deterministic function of ss and public data, the sender can re-derive the spend secret of the output it just created and race the recipient to spend it. Every payment would be clawable; no payment would ever be final. The flaw is structural: the sender must be able to compute the spend *public* key to build the commitment, so any spend key derived from sender-known material is sender-spendable. A recipient-only secret must enter the spend path — and the only place it can live is the address.

Discrete therefore gives each identity a **long-term ML-DSA-65 spend keypair** (Section 4.1), publishes spk in the address, and binds outputs to it via the commitment of Eq. (12). The sender knows spk (it is public) and chooses ρ , so it can still construct the output — but it cannot sign, because it lacks ssk . The resulting properties:

- **Receiving stays private.** An output carries only *spendCommit*; ρ is encrypted and spk is not on chain until a spend. Unspent outputs are unlinkable to any address, and even after one spend reveals spk , the identity’s *other* unspent outputs stay unlinkable (linking them would require their secret ρ values).
- **Spends are publicly linkable.** Spending reveals spk , so all spends by one identity are linkable, and the spent output is identified exactly. This is the deliberate, documented Phase-1 trade-off: the transaction graph is transparent (Section 9).
- **Quantum-safe finality of payment.** Only the holder of ssk — never the sender — can authorize a spend.

5.3 Scanning and wallet recovery

Scanning mirrors the CryptoNote mental model — try every output, and let cryptography sort ownership — with the AEAD tag and the spend commitment as the two gates. Recovery from seed is the same procedure run from genesis.

Two invariants are wallet-critical. First, an output is never credited as spendable until the spend-commitment check passes — AEAD success alone proves delivery, not ownership.

Algorithm 2 SCAN — recipient side, per transaction tx with wallet keys (vsk, spk) .

```

function ScanTx(tx, vsk, spk):
    ih ← inputsHash(tx.inputs)           // from the tx being scanned
    for each (i, out) in tx.outputs where out.type == 0x10:
        ss ← ML-KEM-768.Decaps(vsk, out.kemCt)    // garbage ct ⇒ random ss
        for T known or T = 0:           // recovered T below confirms routing
            ctx ← SHA3-256("discrete-pq-out-context-v1" || ih
                          || out.kemCt || LE32(i) || LE64(T))
            k ← HKDF(ss, "discrete-pq-aead-key-v1" || ctx, 32)
            m ← AEAD.Dec(k, 012, aad=ctx || LE64(out.amount), out.encPayload)
            if m = ⊥: continue           // not ours, or tampered: hard discard
            (rho, T') ← parse m; require T' == T
            if SHA3-256("discrete-pq-spend-commit-v1" || spk || rho)
                ≠ out.spendCommit:
                continue           // ownership gate: never credit without this
            record { txid, i, out.amount, out.unlockHeight, rho, ctx, T }

```

Second, no spend secret is stored per output: the wallet record is $\{txid, index, amount, \rho, context, T\}$, and the long-term ssk is re-derived from the seed on demand.

5.4 Spending, the signing digest, and nullifiers

Signing digest. To spend, the owner constructs the full unsigned transaction and signs a digest that binds the *entire* prefix — version, type, unlock height, every input's outpoint and revealed credentials, every output byte, the tx_extra blob, and the fee:

$$\begin{aligned}
 d = H(& \text{discrete-pq-tx-sign-v1} \parallel ver \parallel txType \parallel \text{LE64}(\text{unlockHeight}) \parallel \text{LE32}(n_{in}) \\
 & \parallel (\text{prevTxid}_j \parallel \text{LE32}(\text{prevOutIndex}_j) \parallel spk_j \parallel \rho_j)_j \parallel \text{LE32}(n_{out}) \\
 & \parallel (\text{type}_i \parallel \text{LE64}(a_i) \parallel \text{LE64}(u_i) \parallel kemCt_i \parallel encPayload_i \parallel spendCommit_i)_i \\
 & \parallel \text{LE32}(|extra|) \parallel extra \parallel \text{LE64}(fee))
 \end{aligned} \tag{13}$$

Each input then carries $\sigma_j = \text{ML-DSA-65.Sign}(ssk_j, d)$. Binding every prefix field that enters the txid closes malleability: a relayer who mutates the type, a lock, the extra field, or any output invalidates every signature.

Nullifiers. Double-spend prevention uses a *nullifier* — a unique, unforgeable marker for the spent output, Discrete's post-quantum analogue of a key image:

$$nf = H(\text{discrete-pq-nullifier-v1} \parallel spk \parallel \rho \parallel prevTxid \parallel \text{LE32}(prevOutIndex)). \tag{14}$$

Nullifiers are **computed by every validating node from revealed fields and never serialized in the transaction** — a node never trusts a provided value, which removes a class of malleability and inconsistent-validation attacks outright. Binding the spent outpoint makes the nullifier unique per output even when two outputs share (spk, ρ) ; this matters because coinbase outputs use a *publicly recomputable* ρ (Section 6.2) that could otherwise be replayed into a colliding output to brick the original. The outpoint is already revealed in the spending input, so nothing extra leaks. The nullifier set is persisted keyed by block height, so a reorg rolls it back exactly (delete at orphaned heights, re-apply along the adopted chain).

Algorithm 3 SPEND — wallet side, spending stored records $R_1..R_n$ to outputs $O_1..O_m$.

```

function Spend(records, outputs, fee, extra):
    (spk, ssk) ← deriveSpendKeys(seedmaster)    // re-derived; never persisted
    for each R in records:
        inputs += { R.prevTxid, R.prevOutIndex, authPub=spk, rhoReveal=R.rho }
    d ← txSigningDigest(ver=1, TX_PQ, unlockHeight, inputs, outputs,
                        extra, fee)              // Eq. (13)
    for each input j:  $\sigma_j \leftarrow \text{ML-DSA-65.Sign(ssk, d)}$ 
    discard ssk from memory
    return SignedTx{ prefix,  $\sigma_1.. \sigma_n$  }

```

Algorithm 4 VALIDATETxPQ — node side, for each TX_PQ transaction.

```

function ValidateTxPQ(tx, chainState):
    require  $1 \leq |\text{inputs}| \leq 32, 1 \leq |\text{outputs}| \leq 64, \text{size}(\text{tx}) \leq 256 \text{ KiB}$ 
    require  $|\text{tx.extra}| \leq 4096$  and  $\text{tx.fee} \geq \text{feeFloor}(|\text{tx.extra}|)$     // Sec. 5.8
    d ← txSigningDigest(tx)      // Eq. (13)
    total_in ← 0
    for each input j in tx.inputs:
        out ← chainState.lookup(j.prevTxid, j.prevOutIndex)
        require out exists, type  $\in \{0x10, 0x11\}$ , mature, height  $\geq \text{out.unlockHeight}$ 
        require SHA3-256("discrete-pq-spend-commit-v1" || j.authPub || j.rhoReveal)
            == out.spendCommit
        nf ← SHA3-256("discrete-pq-nullifier-v1" || j.authPub || j.rhoReveal
                    || j.prevTxid || LE32(j.prevOutIndex))
        require nf  $\notin$  chainState.nullifierSet    // double-spend check
        require ML-DSA-65.Verify(j.authPub, d,  $\sigma_j$ ) == 1
        total_in += out.amount
    for each output i: require wellFormed(i) and canonicalDenomination(i.amount)
    require total_in ==  $\sum_i \text{tx.outputs}[i].\text{amount} + \text{tx.fee}$     // exact balance
    for each input j: chainState.nullifierSet.insert(nfj, height)

```

5.5 Consensus validation rules

Validation (Algorithm 4) is exact: the balance equation admits no tolerance, signatures are verified against the revealed *authPub* that the spend commitment binds, and the fee floor and size caps are consensus rules, not relay policy.

5.6 Canonical denominations

Every output amount must be one of 64 **canonical denominations**: $d \cdot 10^e$ atoms for digit $d \in \{1..9\}$ and decade $e \in \{0..6\}$ (63 values), plus a cap of 10^7 atoms (= 100,000 XDS). Wallets decompose any payment greedily from the largest denomination down; because the smallest denomination is the atomic unit (1 atom = 0.01 XDS), every amount is exactly representable and no dust case exists. Uniform denominations serve two purposes: the on-chain amount universe is identical across wallet implementations (no wallet fingerprinting by split pattern), and the set is *forward-compatible* — the planned confidential-amount design (Section 9) proves membership of a hidden amount in exactly this 64-element table, so outputs created today already live in the right universe and need no migration.

5.7 Per-output time-locks

Each output carries its own *unlockHeight* (0 = none): the output cannot be spent until the chain reaches that height. Because the lock is per-output — distinct from the legacy per-transaction lock — a single transaction can release some funds immediately and vest others over time. The field is bound into the signing digest (13) and the txid, so it is not malleable;

it deliberately does *not* enter (8) or the AEAD associated data, keeping ownership and delivery independent of the lock. The genesis Treasury Reserve (Section 7.5) uses exactly this mechanism, and it is available to any user for vesting and escrow-style payments.

5.8 Fees

PQ transactions are intrinsically large — a spend costs 5,329 bytes per input and 1,193 bytes per output (Table 2). Output count is not a clean proxy for discretionary bloat because the wallet decomposes ordinary amounts into canonical denominations: for example, 1,234 atoms naturally becomes four outputs of 1,000, 200, 30, and 4 atoms. Charging per output would therefore tax the decimal structure of the payment. Discrete retains a predictable **flat base fee**, with a surcharge on the free-form `tx_extra` field:

$$\text{feeFloor}(x) = f_{\min} + \begin{cases} 0, & x \leq 3200 \\ f_{\min} \cdot \lceil (x - 3200)/100 \rceil, & x > 3200 \end{cases} \quad (15)$$

with $f_{\min} = 1 \text{ atom} = 0.01 \text{ XDS}$ and $x = |\text{tx_extra}|$ (capped at 4,096 bytes). The 3,200-byte free allowance is sized so that every protocol-required extra — in particular a paid account-number registration tag of 3,137 bytes — fits without surcharge. The maximum extra surcharge produces a 0.10-XDS floor. Input/output counts and total transaction size remain hard-capped; miners choose which transactions to include; and the dynamic block-size penalty reduces subsidy above the median. A maximal transaction can still pay the same base fee as a small transfer. That is an explicit policy trade-off, not a claim that all size is unavoidable; changing it requires separate economic analysis rather than assuming per-byte pricing is correct.

5.9 Payment proofs

A payer can prove, after the fact and without interaction, that specific outputs of a transaction commit to a recipient’s spending authority. The sender retains each output’s ρ ; the proof artifact

$$(\text{version}=2, \text{genesisId}, \text{txid}, H(\text{domain} \parallel \text{spk}), \{(i, \rho_i)\})$$

lets any verifier recompute Eq. (12) against the claimed spk for each listed output of the canonical transaction and sum the public amounts; forging an entry for a foreign output would require a SHA3-256 second preimage. The proof deliberately does *not* claim ML-KEM delivery: FIPS 203 forbids exposing encapsulation randomness through application interfaces, so no unilateral, non-interactive proof of view-key delivery is possible with the standard API — a fact the design states rather than papers over. Where proof of actual receipt is required, the recipient signs a domain-separated receipt with its spend key (an interactive but fully standard flow reserved as a future wallet surface).

5.10 Transaction sizes

A coinbase output omits the KEM ciphertext and encrypted payload entirely — 1,144 bytes saved per block — because its recipient is recovered deterministically from public data (Section 6.2).

Table 2: Wire sizes (consensus-enforced).

Element	Bytes	Composition
Input (PqInput)	5,329	outpoint 32+4 <i>spk</i> 1952 ρ 32 sig 3309
Output (PqOutput, 0x10)	1,193	amt 8 lock 8 tag 1 <i>kemCt</i> 1088 payload 56 commit 32
Coinbase output (0x11)	49	amount 8 lock 8 tag 1 commit 32
Typical 1-in/2-out transfer	$\approx 7,735$	+ ≈ 20 B envelope
Caps per TX_PQ	—	≤ 32 inputs, ≤ 64 outputs, ≤ 256 KiB

6 Consensus

A proof-of-work chain is secured by hashrate, and hashrate is mercenary: it flows to whatever pays most, and it can be rented, incentivized, or brought to bear against a small chain for the few hours needed to rewrite its history. Discrete’s consensus starts from a principle drawn from that history:

51% resistance cannot come from inside proof-of-work. A unique algorithm, a decentralized pool, or any rule verifiable purely from hashrate can always be satisfied by whoever holds the most hashrate — because hashrate is the only Sybil-resistant resource proof-of-work exposes, and the attacker has a majority of it.

Discrete narrows two attack paths: identity-bound work adds friction to generic delegation, while a local depth limit makes already-connected nodes refuse a late deep rewrite. Neither mechanism removes the underlying majority assumption, and their guarantees are stated separately below.

6.1 Identity-bound signed proof-of-work

Rental markets and pools depend on delegable work. Discrete makes delegation less direct by requiring the reward identity to participate in every attempt, but does not claim that this alone eliminates custodial pools.

Discrete’s proof-of-work is **DiscretePower-1**, a conservative composition of SHAKE-256, ML-DSA-65, and the yespower 1.0 memory-hard core. Let $h(B)$ be the block-hashing blob of block B (major/minor version, timestamp, *prevHash*, nonce, and the transaction Merkle root), and let d_H, d_Q, d_X, d_P, d_F be the five pinned domain strings in Appendix B. The exact construction is

$$H_B = \text{SHAKE256}(d_H \parallel h(B), 32) \quad (16)$$

$$\sigma_B = \text{ML-DSA-65.Sign}(ssk_{\text{miner}}, H_B) \quad (17)$$

$$Q_B = \text{SHAKE256}(d_Q \parallel \sigma_B, 32) \quad (18)$$

$$X_B = \text{SHAKE256}(d_X \parallel H_B \parallel Q_B, 64) \quad (19)$$

$$P = \text{SHAKE256}(d_P, 32) \quad (20)$$

$$Y_B = \text{yespower}_{N=2048, r=32}(X_B; \text{pers} = P) \quad (21)$$

$$\text{PoW}(B) = \text{SHAKE256}(d_F \parallel H_B \parallel Q_B \parallel Y_B, 32) \quad (22)$$

and the block is valid if $\text{PoW}(B) \cdot D \leq 2^{256}$ for current difficulty D , if σ_B verifies against the miner spend public key carried in the coinbase *extra*, and if the coinbase pays that same key (Section 6.2). Because the signature covers the nonce, *every* nonce attempt requires a fresh ML-DSA signature before its yespower evaluation — signing is part of the work

Algorithm 5 MINE — one attempt loop (per thread).

```

function Mine(template B, difficulty D, spend keys (spk, ssk)):
  while not stopped:
    if node is syncing or has no peers: pause // stop-on-isolation, Sec. 6.4
    B.nonce  $\leftarrow$  next nonce
    H  $\leftarrow$  SHAKE256(dH  $\parallel$  hashingBlob(B), 32)
     $\sigma \leftarrow$  ML-DSA-65.Sign(ssk, H) // re-sign EVERY nonce
    Q  $\leftarrow$  SHAKE256(dQ  $\parallel$   $\sigma$ , 32)
    X  $\leftarrow$  SHAKE256(dX  $\parallel$  H  $\parallel$  Q, 64)
    Y  $\leftarrow$  yespower2048,32(X, pers=SHAKE256(dP, 32))
    pow  $\leftarrow$  SHAKE256(dF  $\parallel$  H  $\parallel$  Q  $\parallel$  Y, 32)
    if pow  $\cdot$  D  $\leq 2^{256}$ : submit(B,  $\sigma$ ); break

```

loop, not a one-time setup. The surrounding SHAKE stages define Discrete’s transcript and final target; the underlying memory mixer remains yespower 1.0 rather than a novel lattice-hardness claim.

The construction buys three concrete properties:

- **Reward-identity binding and delegation friction.** A candidate is valid only with a signature from the reward-controlling key, so an unsigned worker cannot redirect the reward and a conventional pool cannot expose one template with a freely variable nonce range. A custodial operator can still sign batches of nonce candidates, distribute signed jobs, accept lower-difficulty shares, and pay workers off-chain. The signing service must scale with aggregate attempt rate; this is friction, not a proof of strong non-outsourcability. Formally strong constructions instead make effective outsourcing give the worker an undetectable way to steal the winning reward [16].
- **ASIC resistance.** yespower’s 8-MiB scratchpad ($N=2048$, $r=32$) is the memory-hardness mechanism, keeping the algorithm CPU-egalitarian. The personalization tag is fixed and Discrete-specific, so work produced for any other chain cannot be reused, precompute-shared, or merge-mine confused with Discrete’s.
- **Tip-binding for free.** *prevHash* is inside the signed, hashed header, so valid work exists only relative to the current chain head; mining requires tracking the tip with no extra machinery.

Discrete states the limits plainly: CPU-oriented memory hardness does not prove permanent ASIC resistance or exclude botnets and farms, and identity binding does not prove that purpose-built custodial pools are impossible.

6.2 Coinbase bound to the block signer

Consensus requires each non-genesis block’s *single* coinbase output to pay the very key that signed the block. The binding uses a *canonical, publicly recomputable* blinding value:

$$\rho_C = H(\text{discrete-coinbase-rho-v1} \parallel \text{spk}_{\text{signer}} \parallel \text{LE32}(\text{height}) \parallel \text{LE32}(i)) \quad (23)$$

$$\text{require: } \text{coinbase.spendCommit} = H(\text{discrete-pq-spend-commit-v1} \parallel \text{spk}_{\text{signer}} \parallel \rho_C) \quad (24)$$

with output index $i = 0$ for normal blocks. Anyone can verify the equality from public data, so *you cannot mine to a key you do not hold*: the identity that performs the work is the identity that is paid. This is an identity-binding property, not one half of a proof of non-

outsourcability. Because ρ_C is public (unlike a normal output's secret ρ), the nullifier's output binding (14) guarantees distinct nullifiers even for outputs sharing (spk, ρ) . Wallets recover mined rewards deterministically — recompute ρ_C for own spk at each height, one constant-time hash comparison per candidate, no KEM decapsulation — which is why coinbase outputs omit the 1,144 bytes of delivery machinery. The genesis block (height 0) is the sole exception to signature validation: it is a fixed candidate constant carrying the Treasury Reserve (Section 7.5) to 21 independent keys, with $i = 0..20$ giving each batch a distinct ρ_C .

6.3 Difficulty adjustment

Discrete retargets every block with **LWMA-1** [11] over a window of $N = 60$ blocks — the algorithm adopted across small proof-of-work chains precisely because it reconverges quickly after hashrate swings. With target spacing $T = 90$ s, solve times t_j (clamped to $[0, 6T]$, out-of-sequence timestamps monotonized) and average difficulty $\bar{D}$ over the window:

$$L = \sum_{j=1}^N j \cdot \min(t_j, 6T), \quad L \geq \frac{N^2 T}{20}, \quad D_{\text{next}} = \frac{\bar{D} \cdot N(N+1) T \cdot 99}{200 L}, \quad (25)$$

i.e. a linearly weighted moving average that gives recent blocks the most weight, with a +5% trim factor (99/200 rather than 1/2) and a floor that caps the downward step. The future-time limit is $7T$ and the median-of-60 past-timestamps rule applies, the standard companions that keep timestamp manipulation unprofitable at this window size.

6.4 Local first-seen depth limit

Identity-bound work does not stop an attacker who funds a majority or operates a signing pool. Discrete additionally uses local observation order: a node can remember the history it already accepted and refuse a late competing branch. This protects that node's prior view; it does not create a globally objective certificate shared across partitions, new nodes, or eclipsed nodes.

The rule. Let $\mathcal{C}$ be a node's current main chain with tip height h , and let $\mathcal{C}'$ be any competing chain whose fork point is at height f . The node applies:

$$\text{adopt } \mathcal{C}' \iff W(\mathcal{C}') > W(\mathcal{C}) \wedge h - f \leq \Phi, \quad \Phi = \text{FINALITY_DEPTH} = 10, \quad (26)$$

with $W(\cdot)$ cumulative work. A reorganization deeper than Φ blocks (≈ 15 minutes at the 90-s target) is **refused regardless of work**. A chain withheld in private and published late — the classic deep reorg and the end state of selfish mining — is rejected by every node on the ground that it was not seen first by that node. Blocks buried Φ deep are locally final relative to that accepted history.

Mandatory rule, subjective input. The depth is a consensus parameter enforced by every node, not a runtime option, but the current chain to which it is applied is local state. Connected nodes that observed the same history make the same decision; partitions or eclipses can produce conflicting locally final histories. An exchange using Φ confirmations must still monitor independent peers and tip freshness.

The stated cost, bounded. A node that continued *producing* blocks while isolated for longer than Φ blocks — a sustained partition, or a miner mining while disconnected — can wedge onto a minority fork that its own finality rule then refuses to abandon. Discrete bounds this cost in three layers:

1. **Scope.** The wedge affects only nodes that were both isolated *and* actively extending their own fork past Φ . A node that merely went offline holds a prefix of the true chain and syncs forward normally; divergence shallower than Φ self-heals by ordinary reorg, accelerated by LWMA retargeting.
2. **Prevention in code.** The built-in miner pauses when the node is syncing and stops outright when all peers disconnect (Algorithm 5), so the most common wedge source — the lone disconnected miner — waits and resyncs cleanly instead of building a divergent fork. This turns an operator-discipline recommendation into a code-enforced guarantee for total disconnection in the reference miner; it does not detect every partition or eclipse that retains stale peers.
3. **First-class recovery.** On refusing a finality-violating reorg, the node raises a persistent `finality_fork_warning` exposing the local and competing tips, the divergence height, and the *peer split* (how many peers offer the competing chain — a human diagnostic, never an input to validity). The operator confirms the majority chain out-of-band (block explorer, independent nodes) and runs a single guarded command, `resync_to_majority -confirm`, which pops the local chain to just below the divergence height and resyncs. The command refuses to run unless the warning is set, so it cannot be abused to force an ordinary deep reorg by hand. Recovery is deliberate and operator-confirmed, never automatic. The out-of-band majority check is a weak-subjectivity assumption and an operator trust boundary, not a pure on-chain validity rule.

The trade is explicit: refusal protects a connected node's already-observed history, while a partition can produce conflicting locally final views that require social recovery. The procedure is recoverable, but consensus alone cannot prove which partition is the majority; exchanges must pause operations until independent observations agree.

6.5 Why the reference implementation includes no decentralized pool

Decentralized pooling (p2pool and relatives) solves *payout variance*, and solves it well — but a pool of any kind works by sharing each block's reward across recent contributors, and sharing work *is* delegating work: it is precisely what SPoW forbids, and the two rules compete for the same coinbase. Nor can a share-chain be rescued by consensus membership rules or by applying first-seen finality to it: a membership rule is still verifiable only from hashrate (a majority attacker runs their own share-chain), and first-seen finality only rejects history that arrives *late* — an attacker who mines a hostile share-chain openly and continuously, as the 2025 attacker did, is first-seen-valid at every height. The empirical record is pointed: the one major chain that relied on decentralized pooling for mining decentralization is the one that was deeply reorganized in 2025, through exactly the aggregation door SPoW keeps shut.

Discrete therefore manages variance on the solo side: a short 90-second block interval keeps solo payouts on a human timescale, and the reference wallet surfaces accumulated mining progress between blocks. This is a weaker variance tool than a shared pool, accepted knowingly — the security it preserves is ranked above the convenience it costs.

6.6 Network layer

Transactions propagate with **Dandelion++** [12] stem-phase routing (epoch 600s, 2 stems, stem probability 0.9, embargo timeout 173s), narrowing the network-layer link between a transaction and its origin IP. Default ports are 9330 (P2P) and 9331 (RPC). An optional DNS-checkpoint channel (`checkpoints.discrete.cash`) lets maintainers publish signed checkpoints; records are verified against a compiled-in allowlist of *post-quantum* signer addresses (ML-DSA-65 over a domain-separated digest, any-of- N so signers can rotate without emergency releases) and the mechanism fails closed — it ships disabled until a signer is provisioned, and checkpoints can only ever *confirm* history within the finality rule, never override it.

7 Monetary policy

7.1 The emission function

Discrete's issuance is a pure function of consensus state, recomputed and enforced exactly by every node. Let A be the total atoms in circulation after the parent block (*alreadyGeneratedCoins*), $S = 2,100,000,000$ atoms ($= 21,000,000$ XDS) the exponential-curve target, $k = 18$ the emission speed factor, and $B = 350,400$ the expected blocks per year (960/day at the 90-s target). The base block reward is

$$R(A) = \max\left(\underbrace{\lfloor (S - A) \cdot 2^{-k} \rfloor}_{\text{exponential phase}}, \underbrace{\left\lfloor \frac{0.02 A}{B} \right\rfloor}_{\text{perpetual tail}}\right), \quad (27)$$

(with the exponential term floored at a vestigial constant once $A \geq S$; in practice the tail overtakes it long before that point). The coinbase must equal $R(A)$ plus collected fees, adjusted by the block-size penalty, *exactly*: a block that mints one atom too many **or too few** is rejected. There is no discretionary and no hidden issuance.

7.2 Two phases

Exponential phase. Early on, each block mints $2^{-18} \approx 1/262,144$ of the remaining distance to S . The initial reward is $\lfloor S/2^{18} \rfloor = 8,010$ atoms $= 80.10$ XDS. Supply follows $A(t) \approx S(1 - e^{-t/\tau})$ with time constant $\tau = 2^{18}$ blocks ≈ 273 days, so the reward decays smoothly — halving roughly every $2^{18} \ln 2 \approx 181,700$ blocks (≈ 6.2 months) — with no cliff-edge halvings.

Perpetual tail. The tail term — Milton Friedman's k -percent rule [13] applied at 2% per year — overtakes the decaying exponential when $A \approx 0.9853 S \approx 20.69$ million XDS, roughly 3.2 years after launch, at a per-block reward of ≈ 1.19 XDS. From then on the reward is 2% of an ever-growing base spread over the year's blocks: supply grows without bound at a long-run rate of $\sim 2\%$ per annum, and the *rate* is what is fixed. S is therefore explicitly **not a supply ceiling**; it is the parameter that shapes the early curve.

7.3 Why a perpetual tail

- **It funds security forever.** There is always a block subsidy, so miner revenue never collapses to fees alone — avoiding the long-term security-budget problem hard-capped

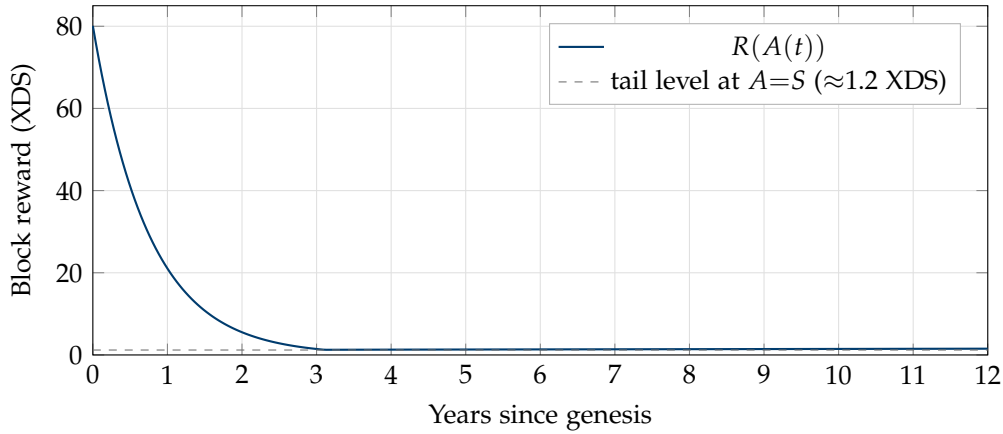


Figure 1: Per-block reward over time: smooth exponential decay from 80.10 XDS (reward half-life $\approx$ 6 months) into the perpetual $\sim 2\%/yr$ tail, which crosses over at $\approx 20.69M$ XDS supply (≈ 3.2 years) and thereafter grows slowly with the base. No halving cliffs; no terminal subsidy collapse.

chains face as subsidies vanish, which for a chain whose whole premise is surviving the *next* decades would be a self-inflicted wound.

- **It targets stable, low, predictable inflation.** A fixed 2% of supply is a shrinking real burden relative to economic activity over time, approximating a constant money-growth rule rather than engineered scarcity followed by engineered famine.
- **It is fully auditable.** $R(A)$ is a pure function of consensus state; anyone can recompute the exact emission at any height.

7.4 Units and parameters

Units	2 decimals; 1 XDS = 100 atoms
Block target	90 seconds
Exponential-curve target S	21,000,000 XDS (shapes the curve; not a cap)
Emission speed factor k	18
Initial block reward	80.10 XDS
Long-run tail	$\sim 2\%/year$, perpetual (≈ 1.2 XDS/block at crossover)

7.5 The Treasury Reserve: transparent and vested

The genesis block allocates a **Treasury Reserve of 1,050,000 XDS — 5% of the exponential-curve target** — to fund development, audits, infrastructure, exchange listings, documentation, and grants. Its terms are enforced by the chain, not by promise:

- The reserve counts *within* the emission target: it reduces the ongoing block reward rather than being minted on top. **There is no ICO, no private sale, and no separate developer tax.**
- It is not liquid at launch. It vests as **21 batches of 50,000 XDS**, one unlocking each quarter over five years, using the per-output time-lock of Section 5.7: batch k (1-based) unlocks at height $(k-1) \times 87,600$ ($\approx$ one quarter of 90-s blocks), so batch 1 is spendable at genesis and batch 21 at height 1,752,000 ≈ 5.0 years.
- The 21 recipients are independent single-key post-quantum accounts held in cold storage; their spend public keys are committed in the candidate genesis constant, so no node or peer can alter the allocation. Each stripped `CoinbaseOutput` stores

`spendCommit( $spk_i$ , coinbaseRho( $spk_i$ , 0,  $i$ ))` with no KEM ciphertext; anyone can re-generate and verify the bytes.

- The genesis `coinbaseextra` embeds a dated headline — “Reuters 08/Jul/2026 — Crypto firms prepare defenses as quantum threat to encryption draws nearer” — Discrete’s analogue of Bitcoin’s Chancellor headline and a statement of purpose. The fixed block timestamp is 22 June 2026, earlier than this headline. Neither timestamp nor headline proves the public announcement time or absence of private mining; the headline only lower-bounds when the candidate artifact could have been assembled.

This is a deliberate, documented departure from a zero-premine model: a young chain needs cryptographic audits and infrastructure, and a transparent, capped, on-chain-vested reserve funds that without hidden issuance or discretionary control.

8 Security analysis

8.1 Threat model

Discrete defends against a network adversary who controls message timing and a minority of nodes; a mining adversary who can rent, incentivize, or own hashrate up to and beyond a majority; malicious transaction counterparties (senders and recipients); and a *quantum-capable archival adversary* who records the entire chain today and later gains a cryptographically relevant quantum computer. Against the last, Discrete’s ownership rests on the Module-LWE/Module-SIS assumptions underlying ML-KEM-768 and ML-DSA-65 at Category 3; symmetric primitives (SHA3-256, ChaCha20-Poly1305) retain comfortable margins against Grover-type speedups, whose quadratic advantage is further blunted in practice by memory-hardness in the PoW path. Discrete does not defend against: compromise of a user’s seed or endpoint; a global network adversary performing traffic correlation beyond what Dandelion++ mitigates; or hardware-scarcity attacks on CPU mining (botnets, large farms), which are stated openly as outside the design’s reach.

8.2 Ownership and supply integrity

- **Spend unforgeability** reduces to ML-DSA-65 existential unforgeability: authorizing a spend requires a signature over the full digest (13) under the exact public key bound into the output’s commitment (12). Substituting a different key requires a SHA3-256 second preimage.
- **Payment finality against the sender.** The sender can construct but never spend an output (Section 5.2) — resolving the clawback flaw in the ancestor design and making receipt final on confirmation.
- **Double-spend prevention** reduces to the collision resistance of the nullifier hash (14) and exactness of the persisted nullifier set (height-keyed, reorg-rolled). Nullifiers are node-derived and never on the wire, so no malleated value can desynchronize validators.
- **Supply exactness.** Every block’s coinbase is checked to equal $R(A) + \text{fees}$ exactly (27); a one-atom deviation in either direction is invalid. Emission cannot be inflated by any party, including miners.
- **Output integrity in flight.** Amount, context, and routing index are bound into the AEAD tag; any mutation makes the output invisible to its recipient rather than mis-

Table 3: Transaction-layer attacks and dispositions.

Attack	Node detects	Wallet detects	Disposition
Malformed field sizes	yes, at admission	—	consensus rejects; all blob sizes fixed
Garbage output (valid sizes)	no	yes (AEAD tag)	inherent to stealth systems; sender burns own coins; recipient unaffected
Amount tampered in flight	no	yes (AAD)	hard discard at scan; recipient unaffected
KEM-ct replay across txs	yes	—	<i>outContext</i> binds inputs hash; same ct in another tx yields different credentials
ρ reuse	not a threat	—	fresh KEM ct $\Rightarrow$ unique context; outpoint-bound nullifier stays unique
Nullifier precollision	infeasible	—	SHA3-256 preimage resistance
Tx malleation by relayer	yes	—	digest (13) binds the full prefix; signatures break on any mutation
Registration squatting	rate-limited	—	first-reg-wins + memory-hard PoW + per-block and mem-pool caps

credited. Output-credential uniqueness is consensus-enforced through the inputs hash (8).

8.3 Attack catalogue

8.4 Consensus attacks

Hashrate rental / hostile pool aggregation. Generic unsigned jobs do not work because every nonce needs the reward identity’s signature. A custom custodial pool can nevertheless retain that key, sign candidate jobs at aggregate rate, and pay shares off-chain. The construction adds operational friction and a signing bottleneck; it does not eliminate aggregation.

Self-funded majority, withheld chain. An attacker who owns a majority and mines privately produces a heavier chain. A connected node that already accepted the honest history refuses it at depth $> \Phi$. New, offline, partitioned, or eclipsed nodes may hold a different local view, and a majority can censor new blocks while it persists.

Selfish mining. Withholding beyond Φ is rejected by nodes that already observed the competing history; within Φ , ordinary cumulative-work selection applies. Profitability is not proved here and depends on propagation, hashrate, and local observation.

Timestamp and difficulty games. Solve-time clamping to $[0, 6T]$, monotonization, the $N^2T/20$ floor on L , and the $7T$ future limit constrain timestamp leverage. Adversarial simulation is still required before launch.

Partition wedges (the accepted residual). A network segment that mines in isolation past Φ wedges and must run the recovery procedure of Section 6.4. The reference miner stops at zero peers, but an eclipsed miner with stale peers or a connected minority segment can continue. Operator-confirmed recovery chooses which local history to discard.

8.5 Costs stated plainly

For the reader auditing trade-offs, the complete list of what Discrete *gives up*, each deliberate: no hard supply cap (a 2%/yr tail is the price of a permanent security budget); publicly linkable spends and transparent amounts in Phase 1; a protocol composition that is test-covered but not yet independently audited; CPU-oriented mining that is not botnet- or farm-proof; possible custodial pooling despite identity binding; and weak-subjective finality that can wedge or conflict under partitions and needs operator-guided recovery.

9 Roadmap: staged discretion

Discrete is a pre-launch, post-quantum, transparent-amount reference chain. Privacy refinements are research proposals for separately reviewed and audited forks. Nothing in this section is consensus until it is implemented, independently reviewed, and frozen.

9.1 Confidential amounts

The one planned refinement is *cash-like discretion over magnitudes*: a payer should not reveal their remaining balance to a payee who can currently read the change output. The direction, recorded in the development specifications:

- The output's plaintext amount is replaced by a committed **denomination bundle**: K fixed slots, each a lattice commitment to one member of the canonical 64-denomination set of Section 5.6 (zero-padded to uniform K , so structure leaks no digit count). Commitments are Module-SIS additive,

$$C = \mathbf{A} \mathbf{r} + v \mathbf{u} \pmod{q}, \quad (28)$$

binding under Module-SIS and homomorphic in v , with q chosen larger than any feasible transaction value so modular balance implies integer balance — eliminating the carry-proof machinery whose retrofitted checks produced audit findings in comparable deployed systems.

- Membership of each slot in the denomination table is shown with a lattice **one-out-of-many proof**; balance is one aggregate identity,

$$\sum_{\text{inputs}} C_j - \sum_{\text{outputs}} C_i - \text{fee} \cdot \mathbf{u} = \mathbf{A} \mathbf{r}_\Delta, \quad (29)$$

proven to open to zero hidden value.

- The ownership model is untouched: ML-KEM delivery, ML-DSA spend authority, and nullifier double-spend prevention carry over; the signing digest additionally binds the value-commitment root so commitments cannot be swapped under a valid signature, and the encrypted payload authenticates against it so recipients reject tampered commitments.

The audit focus is supply integrity — denomination membership soundness, balance, overflow bounds, commitment binding, and malleability — and the fork ships only after independent review. The accepted structural cost is quantization: an amount needing more nonzero digits than K slots splits across outputs, so output count can correlate weakly with amount complexity.

9.2 On untraceability

Discrete deliberately does *not* pursue hiding the transaction graph. The flow of funds stays transparent and publicly auditable — a property that keeps the chain compatible with exchange and compliance expectations, and consistent with its identity as post-quantum *cash* rather than an anonymity system. Confidential amounts add discretion over magnitudes only; they do not make the chain untraceable, and that boundary is by design, not a limitation awaiting a later fix.

10 Related work

Bitcoin [7] established proof-of-work consensus and the UTXO model; its ECDSA ownership and its fee-only terminal security budget are the two long-horizon liabilities Discrete addresses. **CryptoNote** [8] introduced stealth addresses, one-time output keys, and key images; Discrete reconstructs precisely this receiving model on standardized lattice primitives, with nullifiers in place of key images. **Karbo** contributed ancestors of two mechanisms Discrete re-evaluates — identity-bound signed PoW and a local first-seen depth limit — along with the KDSK transaction-family proposal whose ownership flaw Section 5.2 documents and repairs. **Zcash** [9] pioneered the nullifier abstraction Discrete adapts to a transparent setting. Among post-quantum chains, **Abelian** and **BTX** deploy lattice ring-confidential constructions (MatRiCT-lineage and SMILE respectively); their production experience — including publicly documented critical findings in serial binding and pool-credit paths — directly informs the conservative, audit-first staging of Discrete’s confidential-amount fork, and their measured proof sizes anchor its engineering estimates. **QRL** ships hash-based signatures (XMSS) from genesis but retains stateful key management and outsourceable mining. Discrete’s distinct position is the combination: NIST-standardized lattice primitives, CryptoNote-style receiving privacy, identity-bound work, and a mandatory local depth limit, in one compact reference implementation. The composition remains subject to independent review.

11 Conclusion

Discrete answers a question that every existing chain will eventually face — *what does a cryptocurrency look like when it takes the quantum adversary seriously from its first block?* — with deliberately conservative choices: NIST-standardized lattice primitives for every ownership key and signature; CryptoNote-style receiving rebuilt without elliptic curves; DiscretePower-1 identity-bound work; a local depth-10 reorganization limit; and an emission schedule that maintains a subsidy. Its costs and residual risks — large transactions, transparent amounts, possible custodial pooling, no supply cap, conflicting partition views, and operator-guided recovery — are part of the protocol assessment, not hidden behind absolute security claims. The reference implementation is pre-launch and requires independent review.

A Candidate network parameters

These constants are implemented pre-launch candidates. They must be benchmarked, independently reviewed, and frozen in a named release before mainnet; changing them after

Item	Value
Coin name / ticker	Discrete / XDS
Decimals	2 (1 XDS = 100 atoms)
Block target	90 s
Proof-of-work	DiscretePower-1: SHAKE-256 + ML-DSA-65 + yespower 1.0, $N=2048$, $r=32$ (8 MiB)
Mining binding	coinbase recipient = block signer (ML-DSA-65), Eq. (23)
Difficulty	LWMA-1, window $N=60$, no cut/lag; future-time limit 7T
Finality depth Φ	10 blocks (≈ 15 min); mandatory, local-history-relative
Coinbase maturity	10 blocks; spendable age 3 blocks
Emission	Eq. (27): $S = 21\text{M XDS}$, $k = 18$, tail $\sim 2\%/yr$ (no cap)
Treasury Reserve	1,050,000 XDS (5%), 21 quarterly batches over 5 years
Signatures	ML-DSA-65 (pk 1,952 B, sig 3,309 B)
Key delivery	ML-KEM-768 (ct 1,088 B); AEAD payload 56 B
Derivation / AEAD	SHA3-256 + HKDF-SHA3-256; ChaCha20-Poly1305 (IETF)
Address format	extended Bech32m-derived; mainnet <code>disc1...</code> , testnet <code>tdisc1...</code>
Amounts	64 canonical denominations (Section 5.6)
Tx limits	≤ 32 in, ≤ 64 out, ≤ 256 KiB, extra $\leq 4,096$ B
Fees	flat 0.01 XDS + extra surcharge, Eq. (15); max 0.10 XDS
Free registration	identity-bound yespower PoW, target <code>0x00007FFFFFFFFFFFFF</code> (2^{17} exp.), window 60, $\leq 100/\text{blk}$
Transaction relay	Dandelion++ (epoch 600 s, 2 stems, $p=0.9$, embargo 173 s)
Default ports	P2P 9330; RPC 9331
Genesis timestamp	2026-07-13 02:39:16 UTC
Genesis hash	3101b248ac612883c47b93fa6a4d9d34020ee2f7541ad17e73c42088748a5219

launch is a hard fork. Block versions 2–8 and transaction type 0x02 are reserved for genuine future forks.

B Domain-separation tags

All tags are ASCII, hashed without a trailing NUL, pinned byte-for-byte by unit tests and published known-answer vectors.

Derivation	Domain string	Mechanism
View-key seed	discrete-pq-view-root-v1	HKDF, $L=64$
Spend-key seed	discrete-pq-spend-root-v1	HKDF, $L=32$
Deposit spend keys	discrete-pq-deposit-spend-v1 LE32(i)	HKDF, $L=32$
Inputs hash	discrete-pq-inputs-hash-v1	SHA3-256
Output context	discrete-pq-out-context-v1	SHA3-256
AEAD key	discrete-pq-aead-key-v1	HKDF, $L=32$
Spend commitment	discrete-pq-spend-commit-v1	SHA3-256
Nullifier	discrete-pq-nullifier-v1	SHA3-256
Signing digest	discrete-pq-tx-sign-v1	SHA3-256
Coinbase ρ_C	discrete-coinbase-rho-v1	SHA3-256
PoW header	DiscretePower/v1/header	SHAKE-256, 32 B
PoW signature	DiscretePower/v1/signature	SHAKE-256, 32 B
PoW memory input	DiscretePower/v1/input	SHAKE-256, 64 B
PoW personalization	DiscretePower/v1/memory	SHAKE-256, 32 B
PoW finalization	DiscretePower/v1/final	SHAKE-256, 32 B
Free registration	discrete-pq-free-reg-pow-v1	yespower preimage
<i>Reserved, Phase 2</i>	discrete-pq-ct-mask-v1	must not be used by v1 code
<i>Reserved, Phase 3</i>	discrete-pq-serial-v1	must not be used by v1/v2 code

C Wire formats

PqOutput (type 0x10), 1193 bytes: <pre> amount : LE64 unlockHeight : LE64 // 0 = none type : 0x10 kemCt : bytes[1088] encPayload : bytes[56] // ct(40)+tag(16) spendCommit : bytes[32]</pre>	CoinbaseOutput (type 0x11), 49 bytes: <pre> amount : LE64 unlockHeight : LE64 type : 0x11 spendCommit : bytes[32]</pre>
PqInput, 5329 bytes: <pre> prevTxid : bytes[32] prevOutIndex : LE32 authPub : bytes[1952] // long-term spk rhoReveal : bytes[32] signature : bytes[3309] // over Eq.(13) // nullifier: node-derived</pre>	Registration tag (tx_extra), 3137 B: <pre> tag : 1 byte viewPub : bytes[1184] spendPub : bytes[1952]</pre>

References

- [1] P. W. Shor. *Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer*. SIAM J. Computing 26(5), 1997.
- [2] National Institute of Standards and Technology. *FIPS 203: Module-Lattice-Based Key-Encapsulation Mechanism Standard*. August 2024.
- [3] National Institute of Standards and Technology. *FIPS 204: Module-Lattice-Based Digital Signature Standard*. August 2024.
- [4] National Institute of Standards and Technology. *FIPS 202: SHA-3 Standard: Permutation-Based Hash and Extendable-Output Functions*. August 2015.
- [5] H. Krawczyk and P. Eronen. *HMAC-based Extract-and-Expand Key Derivation Function (HKDF)*. RFC 5869, 2010.
- [6] Y. Nir and A. Langley. *ChaCha20 and Poly1305 for IETF Protocols*. RFC 8439, 2018.
- [7] S. Nakamoto. *Bitcoin: A Peer-to-Peer Electronic Cash System*. 2008.
- [8] N. van Saberhagen. *CryptoNote v2.0*. 2013.
- [9] E. Ben-Sasson, A. Chiesa, C. Garman, M. Green, I. Miers, E. Tromer, and M. Virza. *Zerocash: Decentralized Anonymous Payments from Bitcoin*. IEEE S&P, 2014.
- [10] A. Peslyak (Solar Designer). *yespower: proof-of-work focused fork of yescrypt*. Openwall, 2018.
- [11] Zawy. *LWMA difficulty algorithm*. github.com/zawy12/difficulty-algorithms, issue 3, 2018.
- [12] G. Fanti, S. B. Venkatakrishnan, S. Bakshi, B. Denby, S. Bhargava, A. Miller, and P. Viswanath. *Dandelion++: Lightweight Cryptocurrency Networking with Formal Anonymity Guarantees*. ACM SIGMETRICS, 2018.
- [13] M. Friedman. *A Program for Monetary Stability*. Fordham University Press, 1960.
- [14] P. Wuille and G. Maxwell. *BIP-350: Bech32m format for v1+ witness addresses*. 2020.
- [15] P. Wuille and G. Maxwell. *BIP-173: Base32 address format for native v0–16 witness outputs*. 2017.
- [16] A. Miller, A. Kosba, J. Katz, and E. Shi. *Nonoutsourcable Scratch-Off Puzzles to Discourage Bitcoin Mining Coalitions*. ACM CCS, 2015.

- [17] Public reporting on the August 2025 Monero hashrate-majority and deep-reorganization incident (Qubic pool).
- [18] V. Lyubashevsky, N. K. Nguyen, and G. Seiler. *SMILE: Set Membership from Ideal Lattices with Applications to Ring Signatures and Confidential Transactions*. CRYPTO 2021.
- [19] M. F. Esgin, R. Steinfeld, J. K. Liu, and D. Liu. *MatRiCT: Efficient, Scalable and Post-Quantum Blockchain Confidential Transactions Protocol*. ACM CCS 2019.
- [20] J. Groth and M. Kohlweiss. *One-out-of-Many Proofs: Or How to Leak a Secret and Spend a Coin*. EUROCRYPT 2015.
- [21] M. Palatinus, P. Rusnak, A. Voisine, and S. Bowe. *BIP-39: Mnemonic code for generating deterministic keys*. 2013.